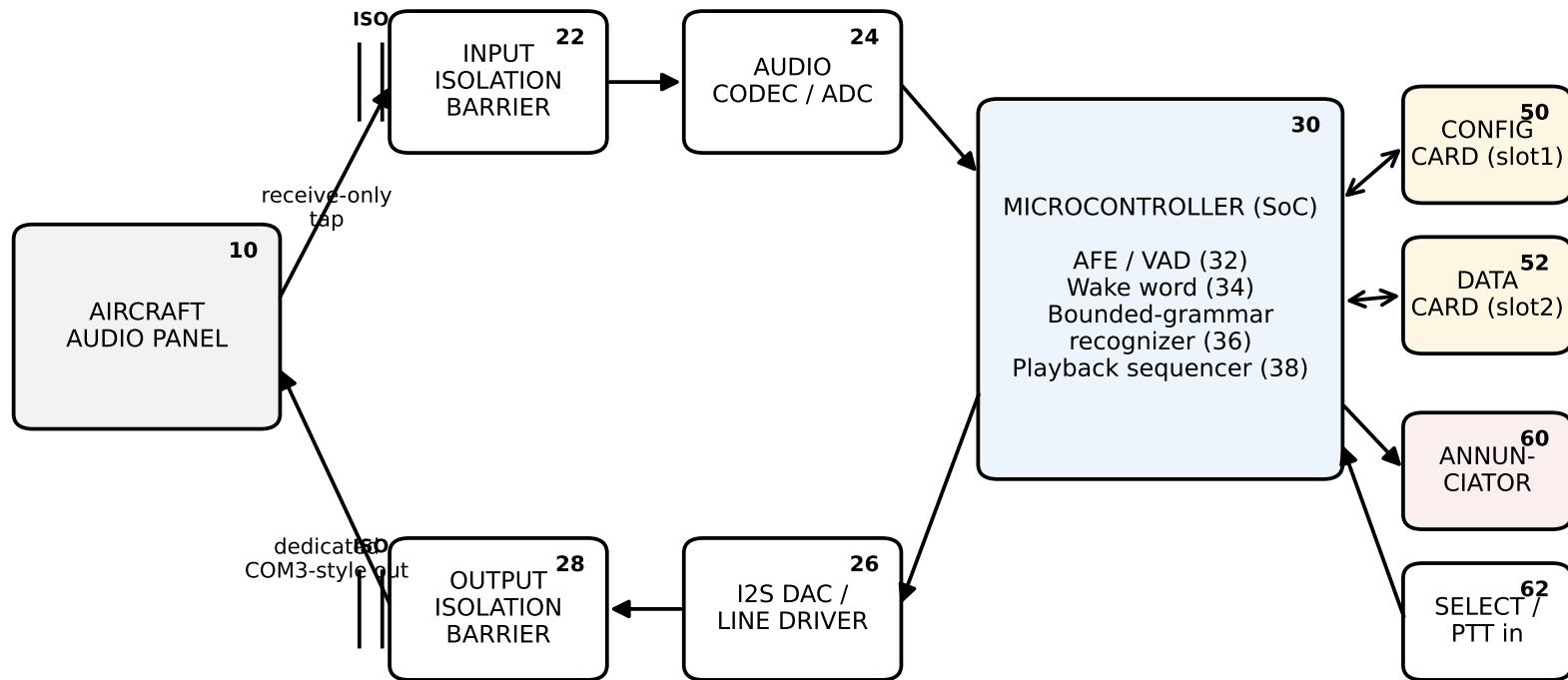


FIG. 1

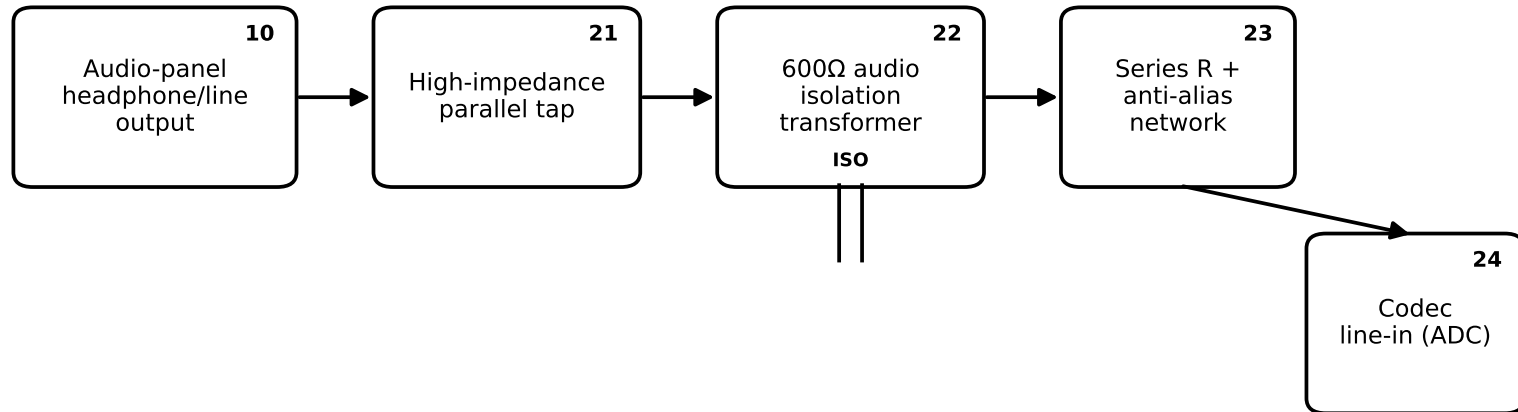
System Block Diagram — Dual Galvanically-Isolated Channels



Two electrically-separate, galvanically-isolated channels: receive-only INPUT tap (22/24) and dedicated isolated OUTPUT (26/28).

FIG. 2

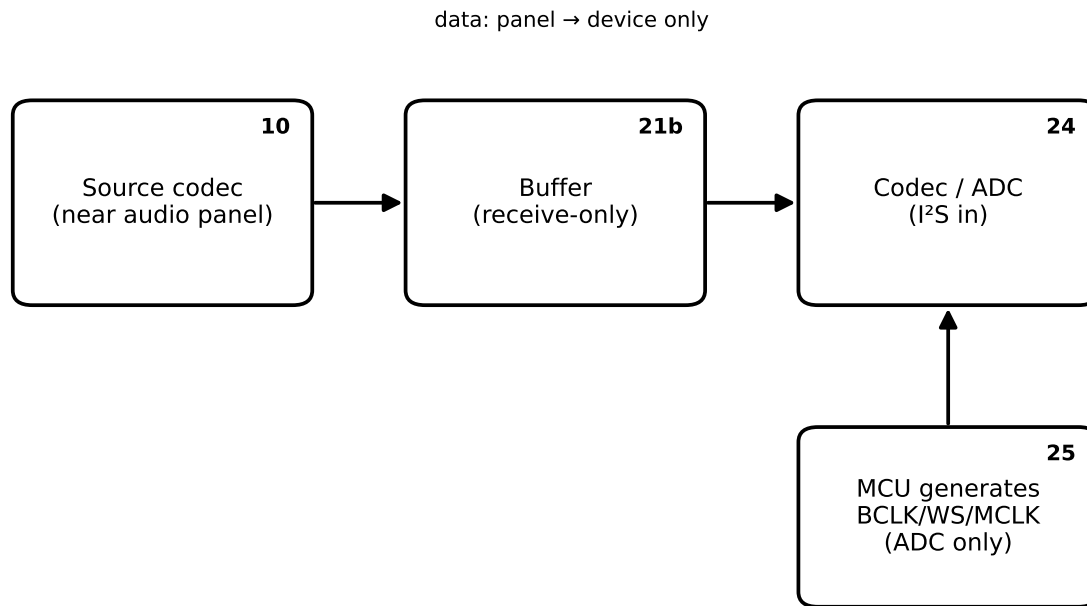
Analog Audio-Input Embodiment (Receive-Only, Isolated)



High-impedance parallel tap loads the panel negligibly; transformer (22) blocks any back-feed via the input path.

FIG. 3

Digital Audio-Input Embodiment (Buffered, Receive-Only)



All clocks generated locally for the ADC; no data line is driven back toward any aircraft bus.

FIG. 4

Dual-Card Configuration Architecture & Cross-Card Check

Installation authority (config) is separated from content authority (data).

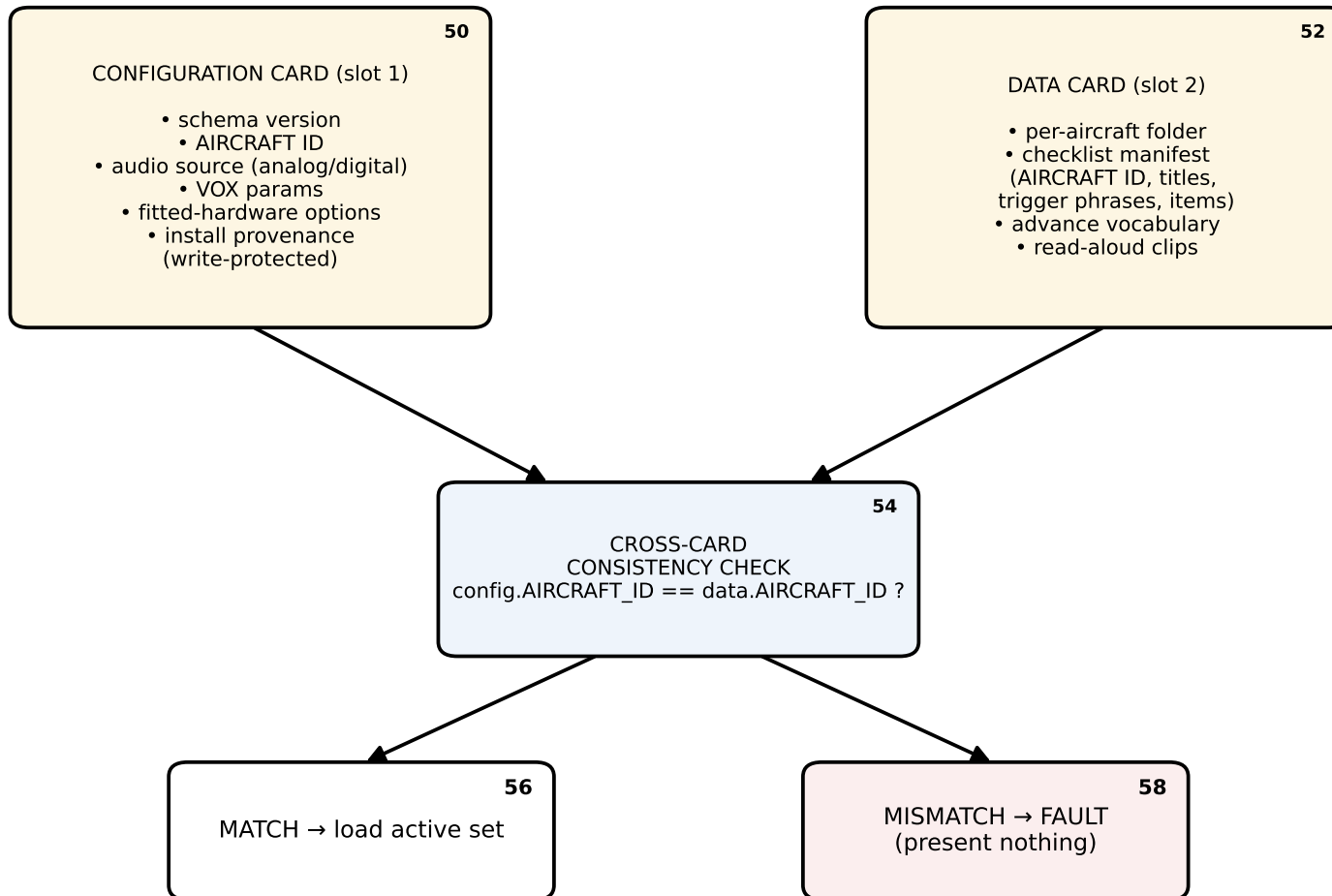


FIG. 5

Boot / Validation State Machine — "Revert-to-Unopened"

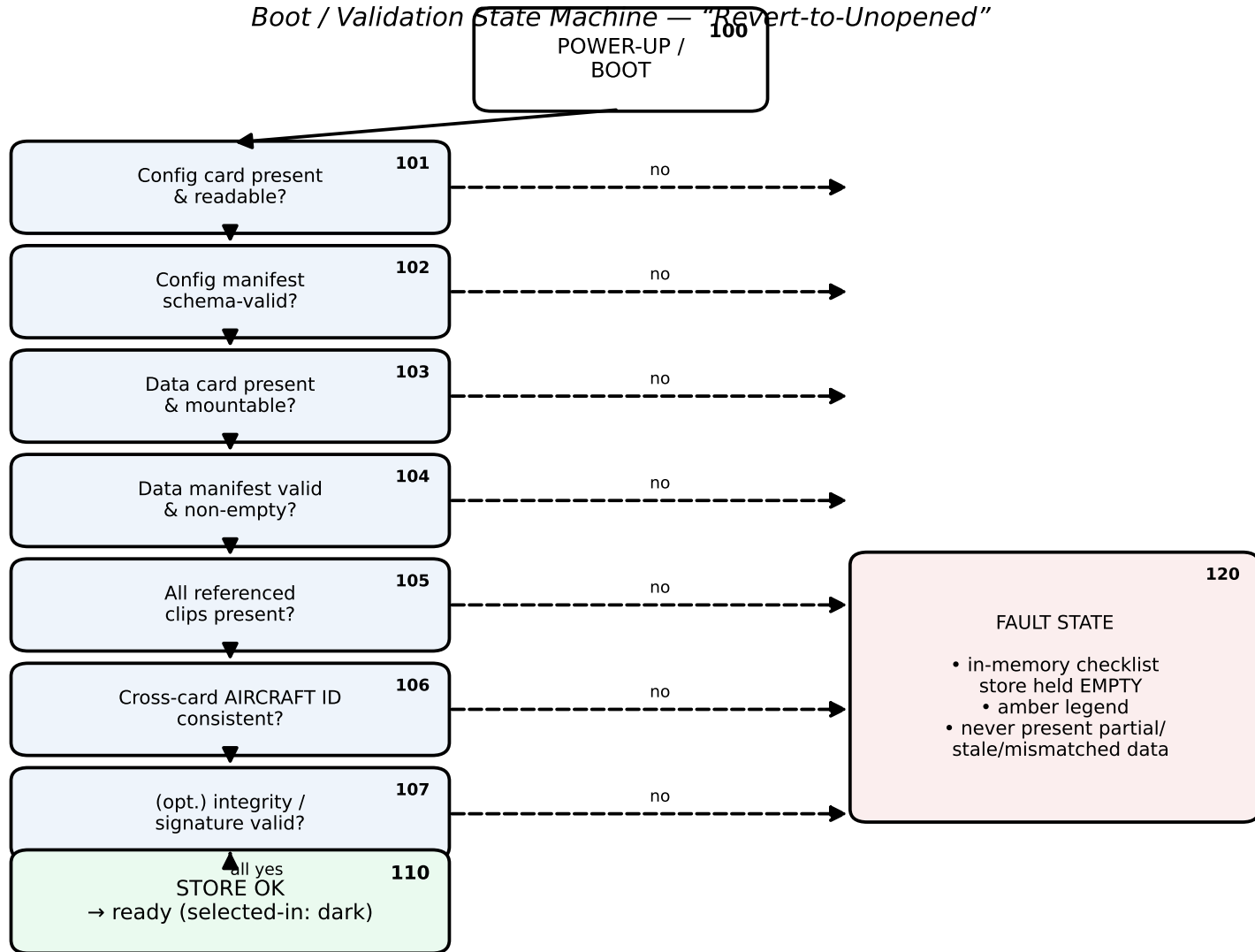


FIG. 6
Dark-Cockpit Annunciator States & Power-Up Lamp Test

CONDITION	ANNUNCIATOR
Selected-IN, healthy	DARK (nothing shown)
Any fault	AMBER 'FAULT' legend
Selected-OUT	WHITE 'OFF'; fault inhibited
Power-up lamp test	All legends ON briefly

A power-up lamp test verifies the indicator so a failed lamp cannot masquerade as a healthy/dark state.

FIG. 7

Interaction Flow — Hands-Free (VOX) with PTT Override

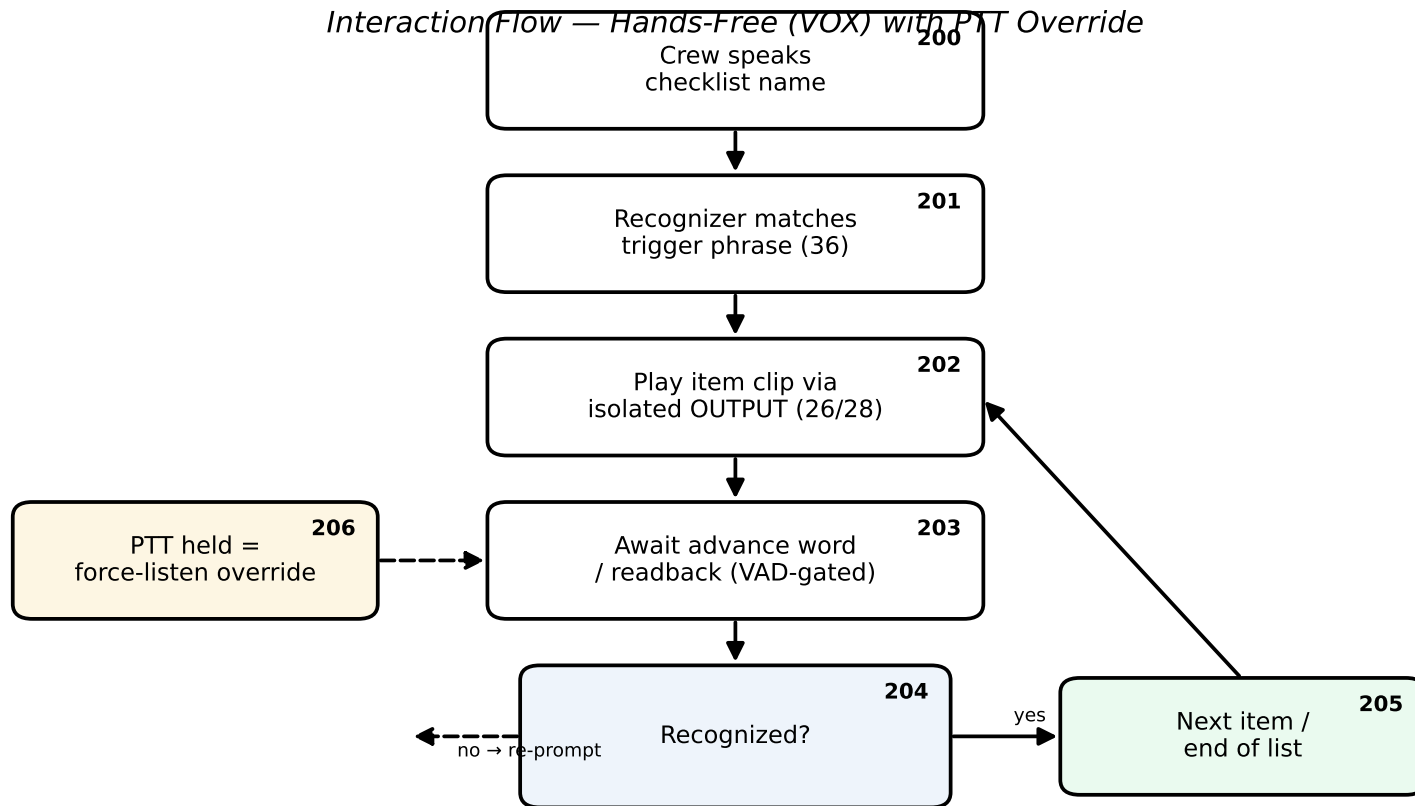
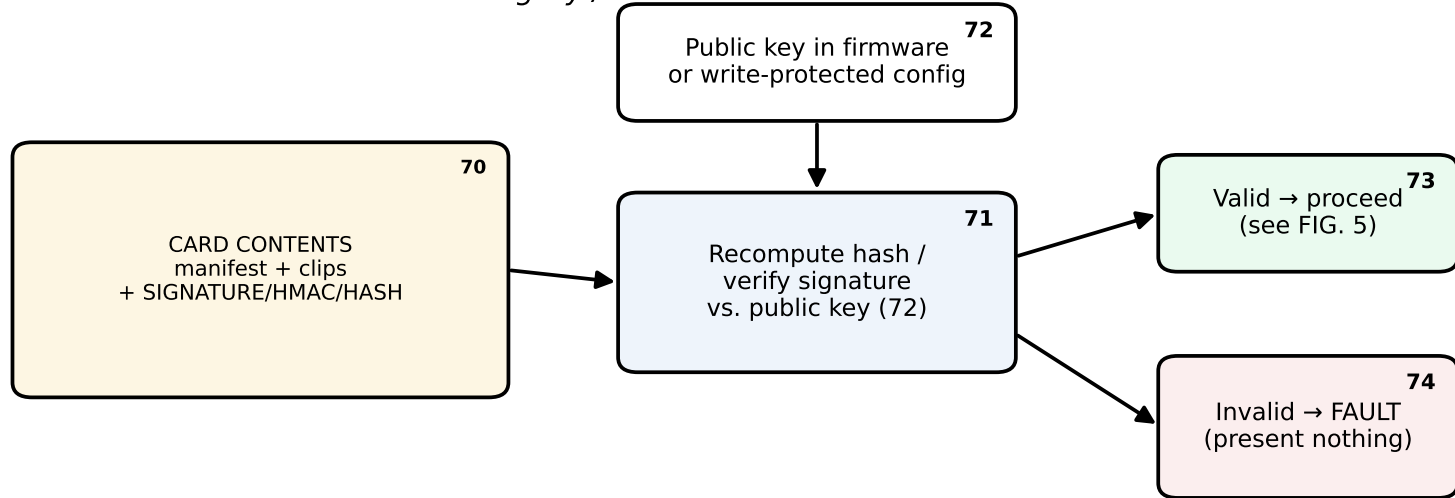


FIG. 8

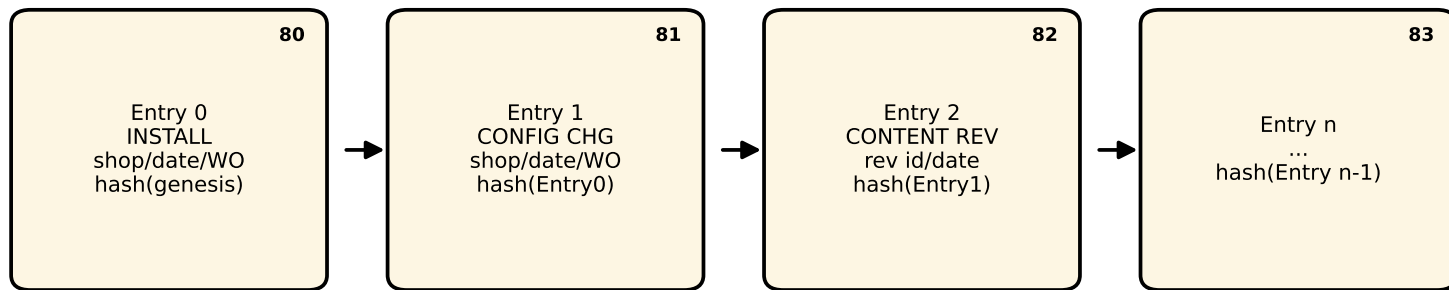
Card Integrity / Authentication Embodiment



Tampered, corrupted, or unauthenticated cards are rejected under the revert-to-unopened rule.

FIG. 9

Tamper-Evident Configuration / Audit Log (Hash-Chained)



Each entry includes a hash over the prior entry; deletion, reordering, or back-dating is detectable by recomputing the chain.

FIG. 10

Multi-Aircraft (Fleet) Data Card with Active-Set Selection

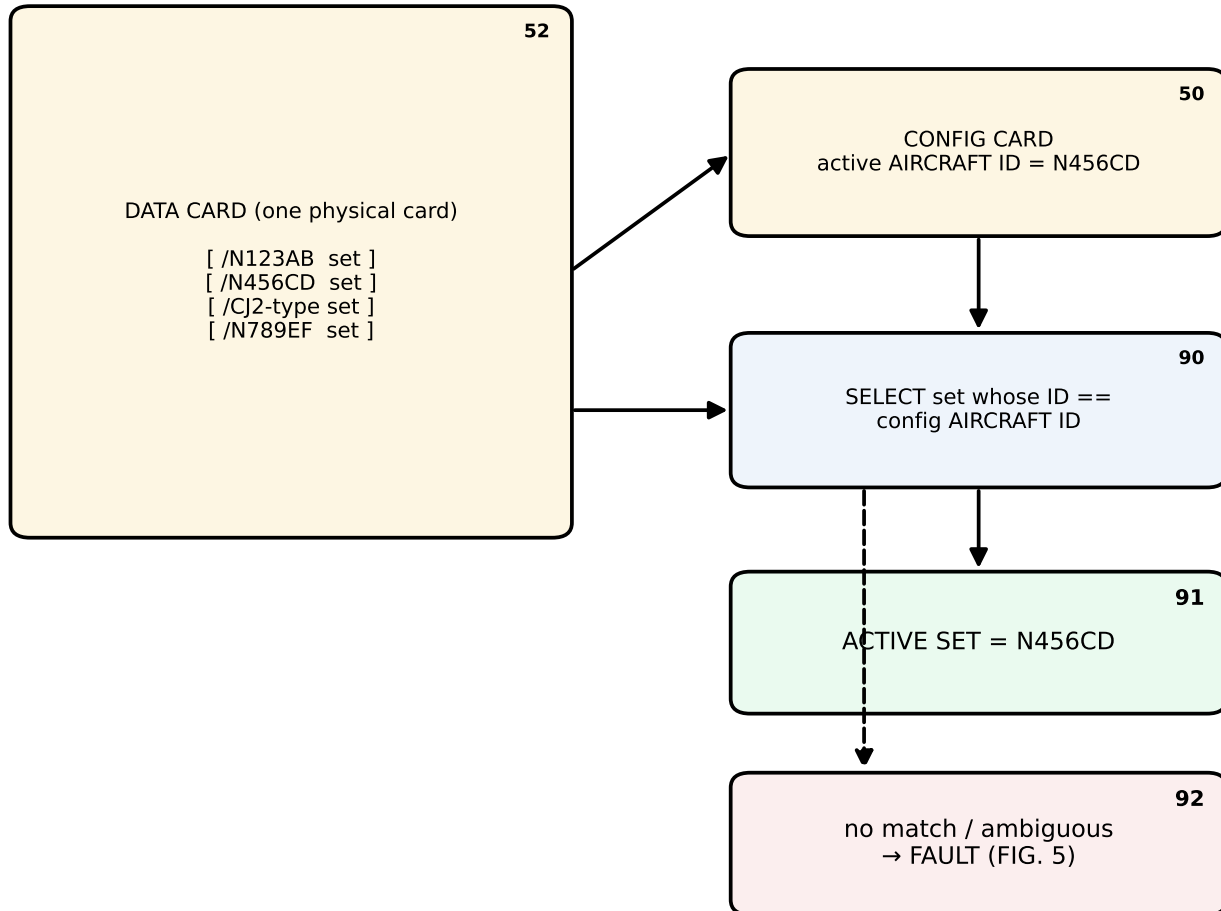
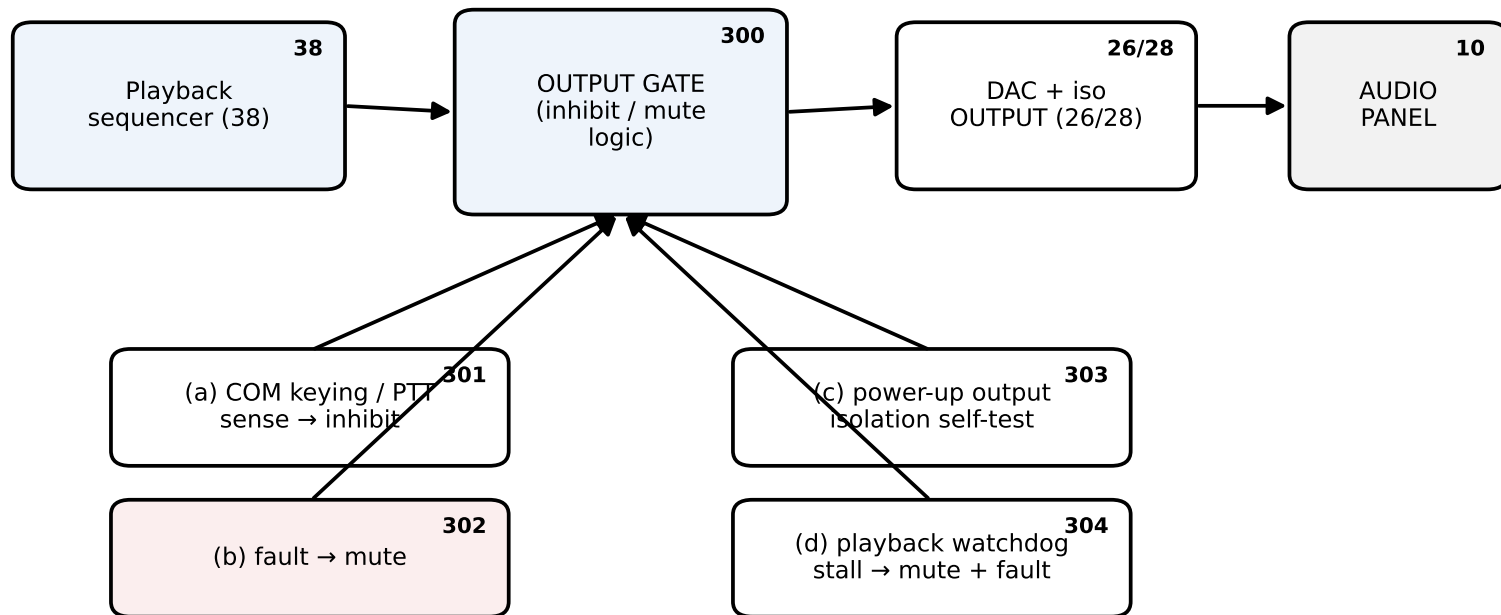


FIG. 11

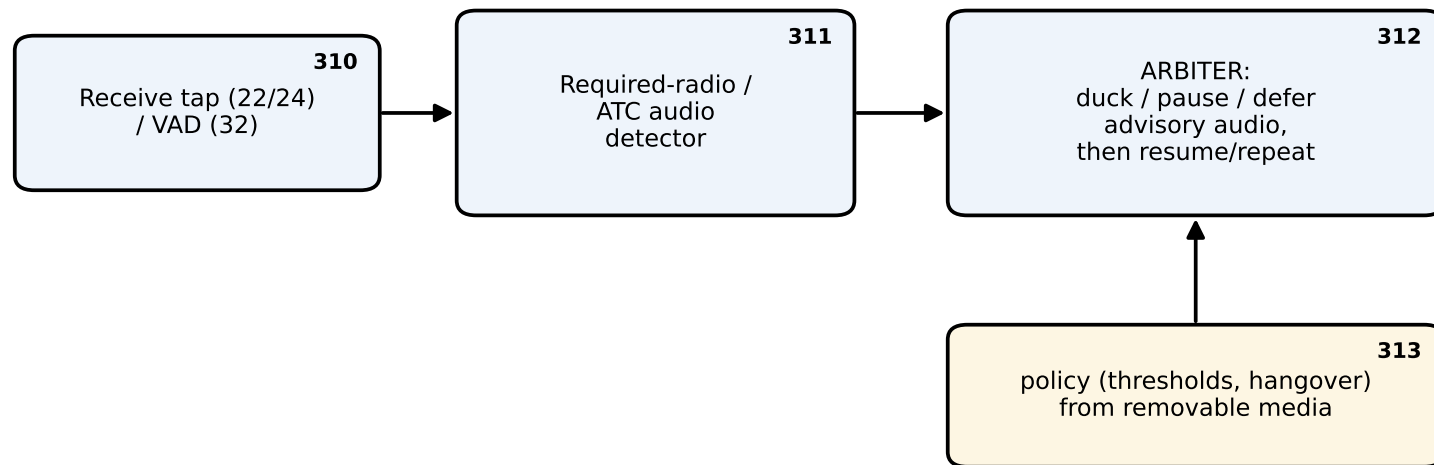
Active Output-Injection Safeguards (Non-Interference)



Non-interference is an actively enforced, testable behavior — not merely a passive consequence of the transformer.

FIG. 12

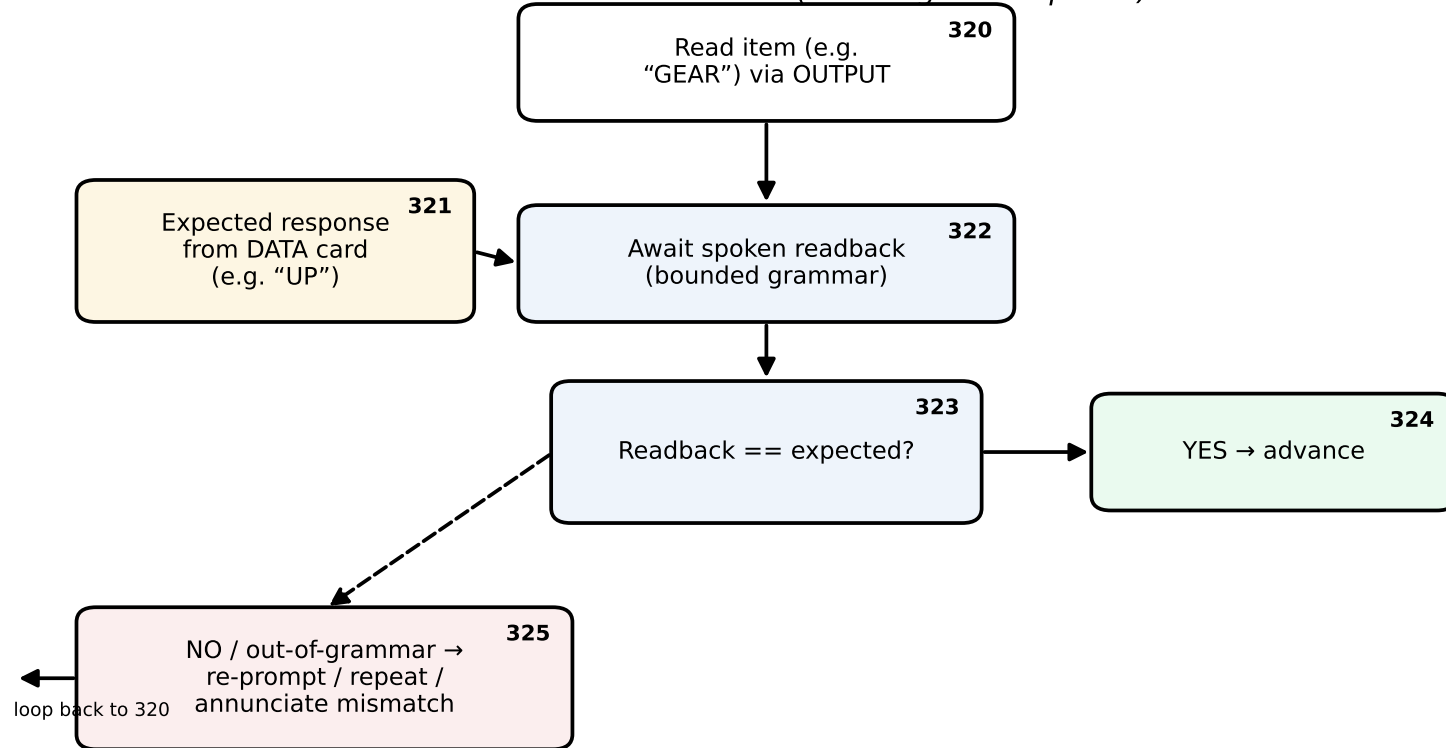
"No-Mask" Audio Arbitration vs. Required Communications



Advisory read-aloud yields to required communications so a checklist item is delivered intelligibly and never talks over ATC.

FIG. 13

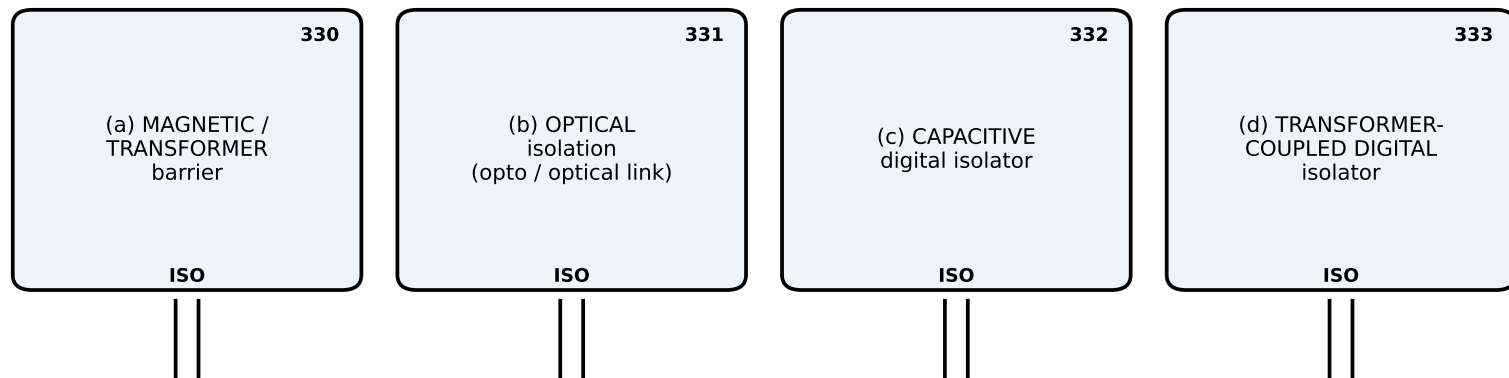
Crew Readback Verification (Challenge & Response)



Items may be configured to accept a generic advance word OR a specific verified readback.

FIG. 14

Interchangeable Isolation-Barrier Embodiments (Either Channel)



Input and output barriers remain INDEPENDENT — no shared conductive path — regardless of which isolation means is used.

Reciting alternatives expressly protects the dual-isolated-channel architecture against substitute-technology design-arounds.